

Embedded Systems & IoT April -July 2026
Lab Exercise: Schematics with ESP32

Part A

1. Calculating the total resistance in ohms

Circuit A

$$\begin{aligned}\text{Total resistance} &= 20 + 30 + 50 \\ &= 100 \text{ ohms}\end{aligned}$$

Circuit B

$$\text{Total resistance} = 1 / R_T$$

$$\begin{aligned}1 / R_T &= 1/R_1 + 1/R_2 + 1/R_3 \\ &= 1/20 + 1/100 + 1/50 \\ &= 0.05 + 0.01 + 0.02 \\ &= 0.08\end{aligned}$$

$$1 / R_T = 0.08$$

$$\begin{aligned}1 / R_T &= 1 / 0.08 \\ &= 12.5\end{aligned}$$

2. Calculating the total current from the power supply

Circuit A

$$\begin{aligned}\text{Total Current} &= \text{Voltage} / \text{Resistance} \\ &= 125 / 100 \\ &= 1.25 \text{ A}\end{aligned}$$

Circuit B

$$\text{Total Current (I}_T\text{)} = I_1 + I_2 + I_3$$

$$I_1 = 125 / 20 = 6.25$$

$$I_2 = 125 / 100 = 1.25$$

$$I_3 = 125 / 50 = 2.5$$

$$\begin{aligned}I_T &= 125 / 10 \\ &= 12.5 \text{ A}\end{aligned}$$

3. Current through each resistor

Circuit A – The current through each resistor is the same – 1.25A

Circuit B - Resistor 1 current = 6.25 A

Resistor 2 current = 1.25 A

Resistor 3 current = 2.5 A

4. The Voltage drop through each resistor

Circuit A

$$V = IR$$

$$R1 = 1.25 * 20 = 25 \text{ V}$$

$$R2 = 1.25 * 30 = 37.5 \text{ V}$$

$$R3 = 1.25 * 50 = 62.5 \text{ V}$$

Circuit B

Voltage is the same across the resistors; therefore, $V = 125 \text{ V}$

5. **Power through each resistor**

Power = Voltage * Current

Circuit A

$$R1 = 25 * 1.25 = 31.25 \text{ W}$$

$$R2 = 37.5 * 1.25 = 46.875 \text{ W}$$

$$R3 = 62.5 * 1.25 = 78.125 \text{ W}$$

Circuit B

$$R1 = 125 * 6.25 = 781.25 \text{ W}$$

$$R2 = 125 * 1.25 = 156.25 \text{ W}$$

$$R3 = 125 * 2.5 = 312.5 \text{ W}$$